

Today, we start working on Calculus-like “things.” The first of which is a limit. Limits will come up repeatedly all year! They are difficult to understand (sometimes), but not difficult to work with.

To illustrate, how do we calculate average velocity? Start by saying $y = f(t)$ and y is a position or distance function. Average velocity is. . .

If the displacement function of a falling object is $y = 16t^2$, find the average speed over the first 2 seconds.

Now for the Calculus part, how do we find the *instantaneous speed* – that is, the speed at any *instant*? Look at the average speed formula:

We want to calculate this as h gets closer and closer to zero (we can’t divide by zero, can we?).

Use the example *from* above to find the instantaneous velocity at $t = 2$ seconds.

Ok, that works well in this case. Let’s write and solve a more difficult one. . .

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

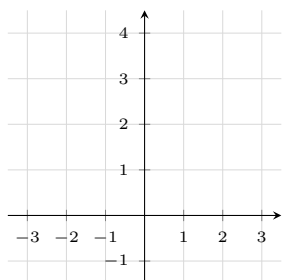
We need a definition (Calculus likes these; it is very theory-based). Assume f is defined in a neighborhood of c and let c and L be real numbers. The function f has the limit L as x approaches c if, given any positive number ε , there is a positive number δ such that for all x :

$$0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon, \text{ we write } \lim_{x \rightarrow c} f(x) = L$$

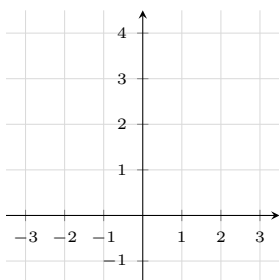
Examples help here. . .

$$\text{Graph } f(x) = \frac{x^2-1}{x-1}, g(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1 \\ 1, & x = 1 \end{cases}, h(x) = x + 1$$

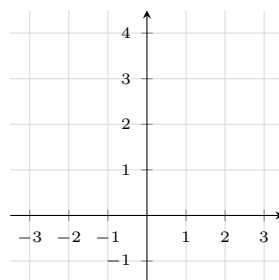
Graph of $f(x)$



Graph of $g(x)$



Graph of $h(x)$



Find the limit for each function as x approaches 1.

We define some properties of limits with L , M , c , and k as real numbers and $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$

- Sum Rule: $\lim_{x \rightarrow c}(f(x) + g(x)) = L + M$
- Difference Rule: $\lim_{x \rightarrow c}(f(x) - g(x)) = L - M$
- Product Rule: $\lim_{x \rightarrow c}(f(x) \times g(x)) = L \times M$
- Constant Multiple Rule: $\lim_{x \rightarrow c}(k \times f(x)) = k \times L$
- Quotient Rule: $\lim_{x \rightarrow c}\left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}, M \neq 0$

Most of these are common sense, but they pop up often. Find the limits:

$$\lim_{x \rightarrow 3} [x^2(2-x)] \qquad \lim_{x \rightarrow 0} \frac{\tan x}{x} \qquad \lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x^2 - 1}$$

It is important to remember that you can only find a limit if it approaches the same thing from *both* sides. See if you can find: $\lim_{x \rightarrow 2} \frac{x^3 - 1}{x - 2}$

This bugs some people, so we say, "Fine, we can also find it from just one side:"

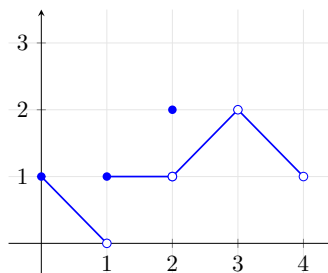
$$\lim_{x \rightarrow 0^-} =$$

$$\lim_{x \rightarrow 0^+} =$$

$$\lim_{x \rightarrow 0} =$$

To illustrate, look at this diagram:

$$\begin{array}{ll} \lim_{x \rightarrow 1^-} f(x) = \underline{\hspace{2cm}} & \lim_{x \rightarrow 1^+} f(x) = \underline{\hspace{2cm}} \\ \lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}} & \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}} \end{array}$$



Finally, something called the (seriously) Sandwich theorem. Don't worry too much about this one (there will be more later).
Show that $\lim_{x \rightarrow 0} [x^2 \sin(\frac{1}{x})] = 0$